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To: Clio Vega (cc Robin Langer)
Title: ($\star$) without BR: the half-integer chain is independent of Prop 4.6
Work-in-progress commit: `d118e27e3e0c724e74070c0fb83dc14989e3eb6c`
Draft: `papers/lg-og-negative-degree/main.pdf` at that commit (27 pp),
`https://github.com/grandpa-rick/work-in-progress/blob/d118e27e3e0c724e74070c0fb83dc14989e3eb6c/papers/lg-og-negative-degree/main.pdf`
Proof file: `proofs/2026-10-27-star-without-BR.md` (same commit). Registry node `q488-oc-star-BR-free`, trust: proved.

Claim

On $LG(m, 2m)$ put $s_x(\kappa) = (-1)^{|\kappa|} H_\kappa(-x)$ for $x \in \mathbb{C}$. The negative-degree recursion ($\star$) (Prop 4.12, `prop:star`)

$$s_{x+1}(\kappa) = s_x(\kappa) + \sum_{\rho \subsetneq \kappa} V(\theta_\rho)\, s_x(\rho) \qquad (x \in \mathbb{C}),$$

where $\theta_\rho = D(\kappa) \setminus D(\rho)$ and $V(\theta)$ is the number of strictly order-preserving maps $\theta \to \{e < o\}$ sending diagonal boxes to e, follows from the LS proof of Thm 1.1 alone. No BR, no (G0), no Prop 4.6 enters.

Proof

Inputs: (1) H_κ is a polynomial (Snapper/Kleiman; also `lem:P` given (G1)), $H_\varnothing \equiv 1$. (2) Thm 1.1(b), proved in §3 from (G1)+(LS)+Ehrhart–Macdonald reciprocity: $s_j(\kappa) = N_\kappa(j)$ for *every* integer $j \geq 1$ and every strict κ with parts $\leq m$ (no κ-dependent lower bound; the zero window is included). (3) Lemma 4.13 (`lem:peel`, peeling the top two values, a pure count): for $j \geq 1$, $N_\kappa(j+1) = N_\kappa(j) + \sum_{\rho \subsetneq \kappa} V(\theta_\rho) N_\rho(j)$.

Fix κ and let $\Delta_\kappa(x) = s_{x+1}(\kappa) - s_x(\kappa) - \sum_{\rho \subsetneq \kappa} V(\theta_\rho) s_x(\rho)$. This is a finite sum of polynomials with integer coefficients independent of x, so a polynomial. For integer $j \geq 1$, (2) applies to every term, and (3) gives $\Delta_\kappa(j) = 0$. Infinitely many zeros, so $\Delta_\kappa \equiv 0$. □

Caveats checked. s_x is defined by the same formula at half-integers, so there is no separate half-integer definition. Thm 1.1(b) and `lem:peel` are both stated for all $j \geq 1$. *Non-circularity.* The draft's second proof of Thm 1.1 uses ($\star$)+`lem:peel`; it must keep the K-theoretic proof of ($\star$). The new proof uses Thm 1.1(b) from §3. Two proofs of ($\star$), run in opposite directions, neither circular.

What changes

Theorem HZ, Cor 1.6 (= Thm B), Prop 6.16(a), Thm Z, and both proofs of the window theorem on LG now depend only on (G1)+(LS). Lemmas 5.1/5.2 and Thm Z are restated with the coefficient $V(\theta)$ of ($\star$): "$V(\theta) \neq 0 \Rightarrow$ no 3-chain" is Lemma 4.4 (`lem:R`), pure combinatorics; the printed chain-drop argument uses only that. Remark `rem:deps` at the start of §6 records this.

Prop 4.6 (BR Cor 5.6 at $p = 1$) is now used only for Thm 4.8 (closed form of the K-theoretic Pieri coefficients c_ρ) and for the second proof of Thm 1.1.

($\star$) *does not give Thm 4.8 back.* Combining it with the hyperplane recursion gives only $\sum_{\rho\subsetneq\kappa}(c_\rho \mp V(\theta_\rho))H_\rho \equiv 0$, and the H_ρ, $\rho \subsetneq \kappa$, are linearly dependent already at $\kappa = (3,1)$: five of them, each of degree ≤ 3. (Census: dependent for $m = 3$ at $(3,1), (3,2), (3,2,1)$; ten shapes at $m = 4$.) So Thm 4.8 genuinely needs K-theory.

Checks

($\star$) checked as a polynomial identity in x against type-C Demazure Hilbert polynomials, V brute-forced from the definition (no BR/Thm 4.8 code), $m \leq 4$ (4, 8, 16 shapes): 0 failures (`scripts/prove258/star_poly.py`). Negative control: adding 1 to any single $V(\theta_\rho)$ breaks ($\star$) at some integer $j \leq m$; all 143/143 perturbations caught ($m = 2$: 6/6, $m = 3$: 27/27, $m = 4$: 110/110).

For the FPSAC cut

The risk item "HZ / Cor 1.6 hinge on Prop 4.6" is gone; only Thm 4.8 still needs the per-result flag. Thm Z is unchanged in content: what you endorsed (uid451) stands; its proof now cites ($\star$) instead of the Pieri support.

Addendum (Dream 258, hunch only)

Why the cheap proof stops at Thm 4.8: ($\star$) is Thm 4.8 (Q30) pushed to a point (χ), and the Hilbert polynomials of the $\rho \subsetneq \kappa$ are dependent, so χ cannot pin down the c_ρ. A T-equivariant (LS-character) version of the peel lemma might recover Thm 4.8 without BR as well. That is for after submission, not for this paper.

Asks. Clio: will you endorse ($\star$)-BR-free, and read `lem:peel` (line-level)?